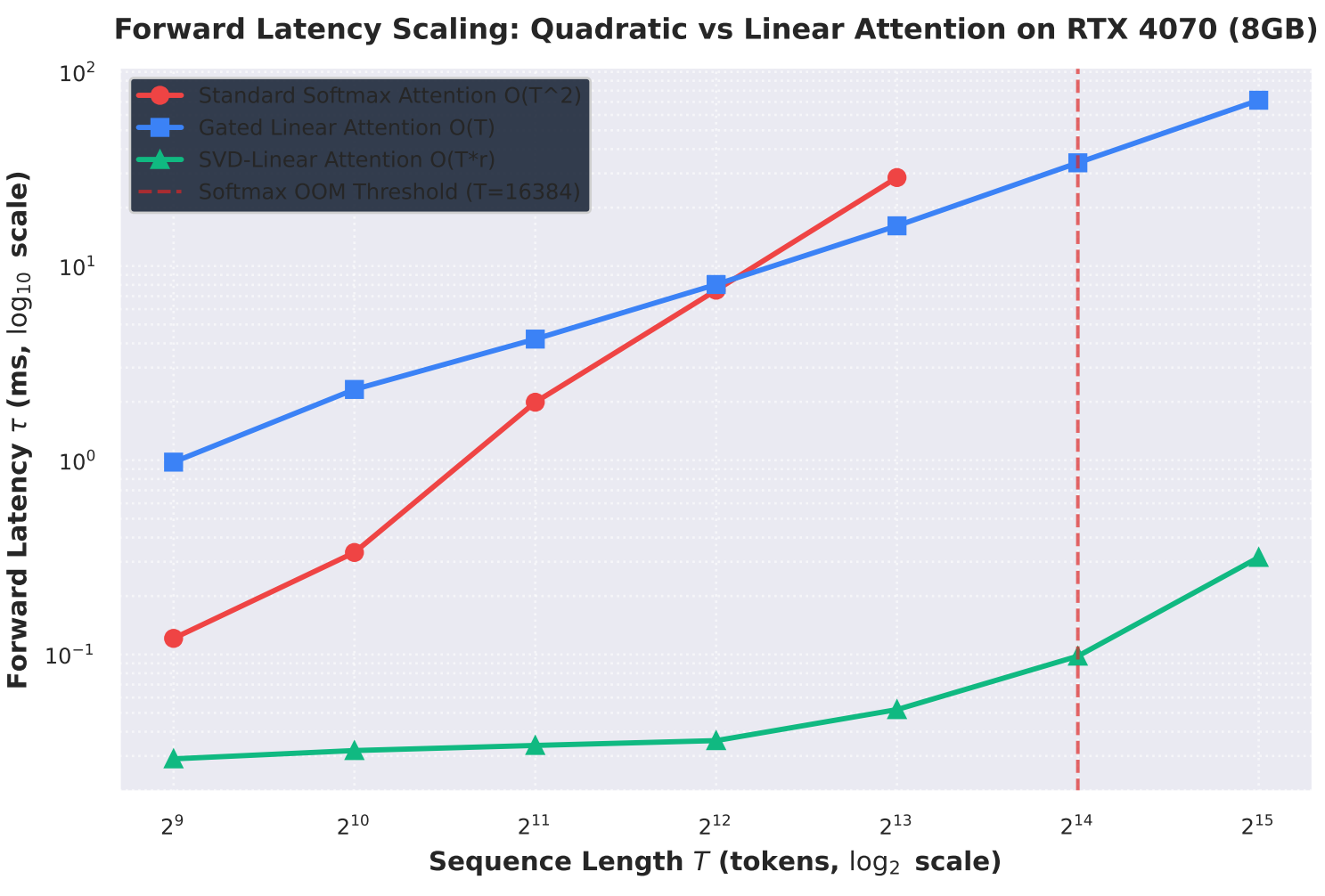
Forward Latency Scaling: Quadratic vs Linear Attention on RTX 4070 (8GB)
Standard Softmax Attention O(T^2)
Gated Linear Attention O(T)
SVD-Linear Attention O(T*r)
Softmax OOM Threshold (T=16384)
Forward Latency τ (ms, log10 scale)
Sequence Length T (tokens, log2 scale)
10^2
10^1
10^0
10^-1
2^9
2^10
2^11
2^12
2^13
2^14
2^15